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E DELL'INFORMAZIONE

CFD Analysis of 3D combustor

LAUREA MAGISTRALE IN AERONAUTICAL ENGINEERING - INGEGNERIA AERONAUTICA

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1. Introduction

The final assignment consists of the CFD analysis of a three-dimensional combustor, by means of the OpenFOAM®-13 software. In particular, the combustion of *n-Heptane* is simulated, injected into the domain as liquid in the form of Lagrangian particles; thus, representing a typical non-premixed combustion problem.

First, a preliminary simulation is performed in order to initiate the flow, which consists of an airflow entering through the inlet at 905 K and 10 m/s . In this precursor case, the Lagrangian spray is deactivated, preventing any chemical reaction and combustion.

The actual problem is then simulated starting from the final time step of the precursor case. Specifically, the fuel is injected ahead of the splitter in the high temperature airflow, leading to fuel evaporation and auto-ignition.

The main goal is to understand the main features of the flow through proper visualizations and by analyzing physical quantities that are representative for the problem at hand.

2. Mesh generation

The geometry of the three-dimensional combustor¹ is shown in Figure 1. In particular, the V-

shaped splitter is positioned inside a cylindrical combustion chamber, while the air flows towards positive values in the y -direction.

In the following, the mesh generation procedure will be analyzed. Notice that the numerical grid here described will be employed for all simulations.

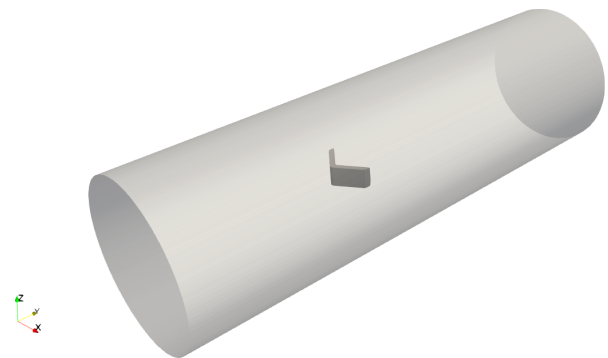


Figure 1: Combustor geometry

2.1. Background mesh

The background mesh is generated by means of the `blockMesh` utility based on the settings specified in `system/blockMeshDict`. The base grid is characterized by a rectangular block of dimensions $0.24 \times 0.8 \times 0.24\text{ m}$, and is discretized by cubic cells with a 24 mm side. The boundary patches are defined as follows: `inlet` for the upstream face, `outlet` for the downstream face, and `outerWalls` for the lateral boundaries.

¹The STL file was provided on the *WeBeep* page of the class.

2.2. Final mesh

The definitive mesh is created using the `snappyHexMesh` utility according to the parameters configured in `system/snappyHexMeshDict`. First, the names of the splitter faces are specified; then, it was decided to increase at 3 the level of refinement through cylindrical elements starting ahead of the splitter and ending behind, as shown in Figure 2. In fact, this represents the area in which the flames spread once the fuel is injected, so the idea was to follow the conical shape of the Lagrangian spray. In the castellated mesh `refinementSurfaces` settings, a refinement level 4 is used for the splitter, and 2 for the `outerWalls`. In fact, since the most critical zone for the flow is in proximity to the splitter and in its wake, it was decided to employ a more refined grid in this region compared to the external walls, where turbulence is expected to be limited. The `resolveFeatureAngle` was set to 10° . Instead, the `snapControls` settings were not modified, while 5 and 3 layers were set over the `outerWalls` and the splitter surfaces respectively. Concerning the layering controls an expansion ratio equal to 1.15 and feature angle 310° were employed.

This tuning allowed to obtain the final mesh shown in Figure 2, which is characterized by about 401k cells and relatively low skewness and non-orthogonality². Clearly, the results of the simulation would improve by extending the refinement region further downstream, allowing the reduction of numerical diffusion in the wake, but this was not possible due to the limit imposed on the cell count.

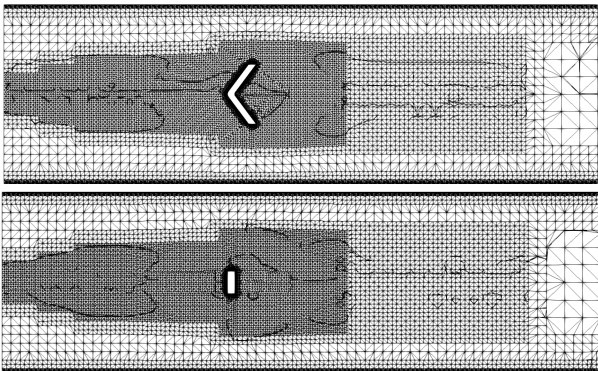


Figure 2: Cuts of the mesh in the xy and zy planes

²`CheckMesh` returned 1.1 of skewness and 64.99° of maximum non-orthogonality.

3. Cold-flow simulation

The precursor simulation allows for the initialization of the flow inside the combustor. In particular, as anticipated in section 1, the spray injection is deactivated; therefore, only high temperature air flows through the domain. It was decided to perform a transient simulation³ lasting 0.05 s, which is able to cope with the unsteadiness of the flow. This solution represents a good compromise between accuracy and computational cost, considering that the goal is to simply perform a preliminary simulation.

3.1. Pre-processing

3.1.1 Boundary conditions

The numerical set-up is based on the cases proposed in the third and fourth assignments and on the `multicomponentFluid/aachenBomb` tutorial case.

First, the boundary conditions were set in the `0.orig` folder. In particular, a 10 m/s velocity was chosen for the `inlet` patch, while a non-slip condition is imposed on the solid walls and `zeroGradient` for the `outlet`. Atmospheric pressure was then considered for the `internalField` with `zeroGradient` for all patches. However, the `waveTransmissive` boundary condition was set for the `outlet`; in fact, if a constant static pressure is set on the domain outlet, negative pressure waves are reflected into the domain, which could lead to numerical instability. Thus, a non-reflective boundary condition allows to overcome such problems by solving the inviscid advection of a wave to simulate an additional domain length.

The air is simplified as constituted by 21% of oxygen and, the remaining 79%, by nitrogen; therefore, it is necessary to include the proper boundary conditions for such chemical species. To this aim, the mass fraction was imposed to 0.766 for `N2` and 0.234 for `O2`. The `Ydefault` is used for initiating the species vector in case not all species defined in the `reaction` file are found. Instead, since combustion is deactivated for this preliminary simulation, the fuel is not injected yet.

Instead, the boundary conditions employed for the temperature and turbulence properties are

³With a maximum CFL number greater than 1.

Patch	T	alphat	k	epsilon	nut
internalField	uniform 300	uniform 0	uniform 0.375	uniform 2	uniform 0
inlet	uniform 905	calculated uniform 0	turbulenceIKEI intensity 0.05 uniform 0.375	turbulenceMLDRI mixingLength 2e-2 uniform 2	calculated uniform 0
outlet	zeroGradient	calculated uniform 0	zeroGradient	zeroGradient	calculated uniform 0
walls	zeroGradient	compressible::alphatWF	kqRWF uniform 0.375	epsilonWF uniform 2	nutkWf uniform 0

Table 1: Boundary conditions

reported in Table 1⁴. In particular, the incoming airflow is characterized by a temperature of 905 K, as anticipated in section 1, and it is notable that the solid walls are considered adiabatic due to the `zeroGradient` boundary condition. *RANS* wall functions are employed for all solid walls.

3.1.2 Numerical set-up

The first-order Euler scheme is employed for time derivatives in `system/fvSchemes`. The `Gauss limitedLinear 1` divergence scheme is used for all variables, except for the compressible stress tensor and the non-linear stress, for which the `Gauss linear` scheme is employed. The `Gauss linear corrected` is then set for the Laplacian schemes.

In `system/fvSolution`, the diagonal solver with $1e-9$ absolute tolerance is used for the density, while for pressure the GAMG solver with the `GaussSeidel` smoother and 0.01 relative tolerance and $1e-6$ absolute tolerance is employed. Then, for the kinetic, turbulence, and thermodynamic variables, the `PBiCGStab` solver with a DILU preconditioner, $1e-6$ absolute tolerance and 0.1 relative tolerance is used. The same for the chemical species as well, but with a null `relTol`.

The simulation was performed in PIMPLE mode with 2 `nCorrectors` and `nNonOrthogonalCorrectors`, 3 `nOuterCorrectors`. Furthermore, it was decided to deactivate the `momentumPredictor` because it allows to avoid the resolution of

the momentum equation linear system, hence reducing the computational time. Since the simulation is run in PIMPLE mode, it was necessary to set the relaxation factors: 0.3 for the `p` field; 0.7 for the `U` and 0.5 for the `k`, `epsilon`, and `h` equations.

Then, in `system/controlDict`, the solver set was the `multicomponentFluid`, even though the Lagrangian spray is still deactivated. The `maxCo` was chosen equal to 5; in fact, as already explained in section 3, the aim is to perform a preliminary simulation to just initialize the flow and such choice allows to significantly limit the computational time, but with an acceptable level of accuracy. A transient simulation cuts down the higher frequency turbulent oscillations, but allows to capture the overall physics of the flow.

The `system/physicalProperties` is required to include the equation of energy; in particular, the `hePsiThermo` model is employed.

3.2. Post-processing

The post-processing representations are produced through both the ParaView software and the OpenFOAM function objects, while the results are mostly processed with MATLAB.

In particular, the residuals were exploited by including the homonymous function object inside the `controlDict`, while the y^+ values were computed through the `yPlus` function object. Instead, the flow visualizations were produced by employing the cutting plane filter in ParaView.

3.3. Convergence

Figure 3 reports the time evolution of the residuals for the main physical variables during the precursor cold-flow simulation. The transport

⁴Where IKEI stands for IntensityKineticEnergyInlet, WF for WallFunction, and MLDRI for MixingLength-DissipationRateInlet, while the `walls` patch refers to all the solid walls.

equations for turbulence and energy show a smooth and monotonic convergence, rapidly settling below 10^{-3} .

Regarding the pressure, the residuals appear to stagnate at relatively high values. This behavior was expected and it represents a direct consequence of the numerical set-up employed: specifically, the deactivation of the `momentumPredictor` combined with the PIMPLE algorithm. The value plotted corresponds to the initial residual at the beginning of the time steps and without the momentum predictor, the solver enters the pressure equation without a preliminary velocity field⁵, leading to a large initial error in the mass conservation (pressure equation). However, this does not imply a lack of convergence; in fact, the solver log confirms that during the inner iterations of the PIMPLE outer loops, the pressure residual gradually drops, ensuring that the pressure-velocity coupling is correctly resolved at every time step. Furthermore, it is worth noting that velocity residuals are not reported in the graph. Since the `momentumPredictor` is disabled, the linear system for the velocity is not solved at the start of the loop and the `residuals` function object, which returns only the initial residuals for each variable, does not output any value for the velocity field.

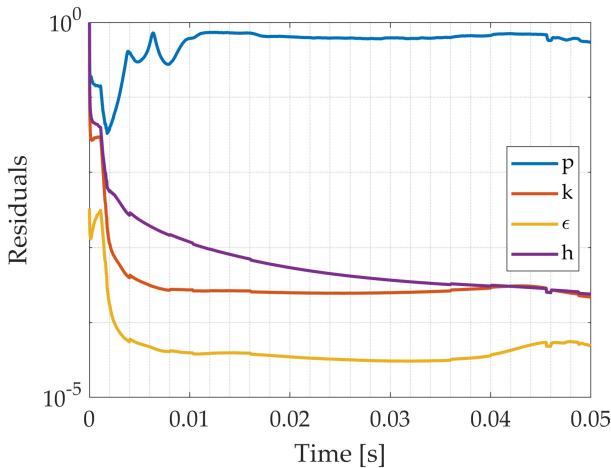


Figure 3: Evolution of the initial residuals over time

⁵The momentum predictor step involves solving the linear system for the velocity using the pressure from the previous step. By skipping it, the preliminary velocity evaluation is avoided.

3.4. Results

3.4.1 Mesh verification

A verification of the mesh was performed by evaluating the distribution of the dimensionless wall distance y^+ during the precursor simulation. The analysis reported a minimum value of 0.36, a maximum of 75.01, both computed over the splitter surface, and a domain-averaged value of approximately 15.85.

These values indicate that the mesh resolution near the walls varies significantly across the domain. The maximum value falls within the logarithmic layer, $30 < y^+ < 300$; this represents the ideal range for the employed *RANS* wall functions, ensuring that the turbulent boundary layer profile is correctly modeled in the high-velocity regions.

Instead, the minimum and the average values suggest that in many areas, probably near the recirculation or stagnation points on the splitter, the mesh is refined enough to resolve the viscous sub-layer. Although standard wall functions are designed for higher y^+ values, the boundary conditions implemented in OpenFOAM are robust and capable of handling these mixed resolution regimes, ensuring a sufficiently accurate wall treatment throughout the domain.

To improve the near-wall grid resolution, an optimization strategy would aim to avoid the immediate buffer layer ($5 < y^+ < 30$), where the current average value falls.

3.4.2 Main flow features

The main flow features resulting from the precursor cold-flow simulation are analyzed in terms of velocity and temperature fields at the final time step ($t = 0.05$ s).

Figure 4 illustrates the velocity magnitude contour on the xy and yz symmetry planes. In particular, the V-shaped splitter acts as a bluff body, dominating the flow behavior.

Upstream of the obstacle, a stagnation region with near-zero velocity is clearly visible. As the flow moves around the object, it is forced to accelerate due to the blockage effect, creating high-velocity regions (orange) at the outer tips of the V-shape. Immediately downstream, a wide wake region is established. This area is characterized by low momentum and flow recirculation, typi-

cal of bluff-body aerodynamics. The separation of the boundary layer from the sharp edges of the splitter generates shear layers that bound this recirculation region.

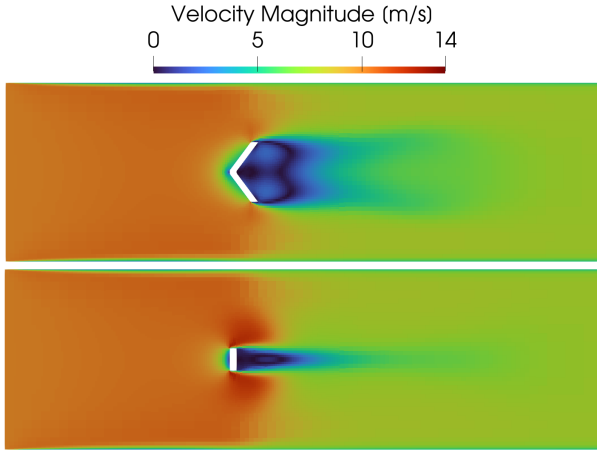


Figure 4: Velocity field at $t = 0.05$ s in the xy and yz symmetry planes

Instead, Figure 5 displays the temperature field at the same instant. Since the simulation is non-reactive, the temperature acts as a scalar variable transported by the flow.

As explained in subsection 3.1.1, the domain was initiated at 300 K (black), while the air enters the inlet at 905 K (yellow). At the final simulation time step, the high-temperature airflow has traveled through most of the combustor, driving out the initial cold air. However, the wake region behind the splitter remains significantly colder, which is physically consistent with the velocity field observed in Figure 4; in fact, the recirculation zone is characterized by low velocities. Hence, the convective transport is much slower in the wake compared to the outer flow, and the hot inlet air has not yet fully penetrated and mixed with the fluid trapped behind the splitter.

This confirms that, although the aerodynamic flow field is almost completely developed, the thermal field is still undergoing a transient process.

4. Hot-flow simulation

The hot-flow simulation is performed starting from the latest time step of the precursor case. Now, the fuel is injected into the domain as liquid in the form of Lagrangian Particles; therefore, since combustion and chemistry are en-

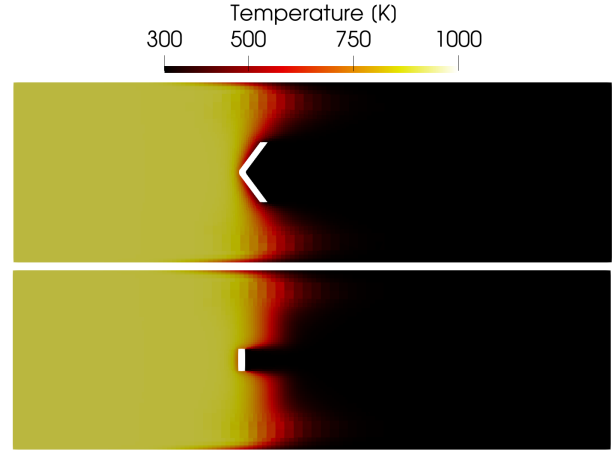


Figure 5: Temperature field at $t = 0.05$ s in the xy and yz symmetry planes

abled, the ignition is caused by the interaction of the spray with the high temperature airflow.

4.1. Pre-processing

The simulation is properly set up taking as starting point the cold-flow precursor case.

4.1.1 Boundary conditions

The boundary conditions employed for the hot-flow simulation are provided by the final time step of the cold-flow case through the `mapFields` utility, which requires including `system/mapFieldsDict`⁶.

4.1.2 Physical set-up

The activation of the combustion and chemistry requires including the `chemkin` folder from the `aachenBomb` tutorial case, which, through the `chemkinToFoam` utility, provides the `reactions` and `speciesThermo` files. These are crucial to simulate the combustion; in fact, the first describes the chemical reaction (1-step) occurring when the fuel enters in contact with the hot-air, while the second one contains all properties of the chemical species involved in such process. The chemistry must be enabled in `constant/chemistryProperties`, where the `seulex` implicit ODE solver is employed with $1e - 8$ absolute tolerance and 0.1 `relTol`. The combustion is then enabled in

⁶Copied from the `incompressibleFluid/hopperEmptying` tutorial case.

`constant/combustionProperties`; in particular, the partially stirred reaction model is used with a `Cmix` coefficient equal to 1. In this model the computational cell is split into two locally uniform zones: one where the reaction takes place, and another one characterized only by mixing. The species transfer occurring between them is driven by a characteristic turbulent mixing time scale.

The Lagrangian spray is modeled by the `constant/cloudProperties`, where the `reactingCloud` model is employed. The injection of 95 mg of *n*-Heptane starts at initial instant⁷ and lasts 0.1 s, while the flow rate profile is characterized by an half-sine function. The injector is located in the longitudinal plane ahead of the splitter at $y = -0.295$ m, and the liquid spray is oriented towards the positive y-direction. 500 thousand parcels per second are injected from a disc geometry, following a conical shape. Then, it was decided to adopt the Reitz-Diwakar breakup model, which is less expensive in terms of computational costs since it is based on simpler semi-empirical relations compared to the Reitz-KHRT model, and is still able to properly reproduce the physics of the Lagrangian spray.

In `constant/physicalProperties`, the fuel employed in the simulation must be specified, which is `C7H16` in liquid phase.

4.1.3 Numerical set-up

The maximum Courant number was lowered to 0.3 in the `controlDict`; in fact, the ODEs that must be solved to couple the chemistry to the fluid dynamics are very "stiff" to converge, so it is necessary to significantly limit the time step advancement.

Then, the `turbOnFinalIterOnly` option was deactivated to obtain a more accurate physics. This was not done in the precursor simulation to reduce the computational cost, since that represented just a preliminary analysis for which it was not required particular accuracy.

4.2. Post processing

The post-processing representations are produced through both the ParaView software and the OpenFOAM function objects, while the re-

sults are mostly processed with MATLAB.

In particular, the average pressure and temperature at the `inlet` and `outlet` patches was computed by employing the `surfaceFieldValue` function object and specifying the `areaAverage` operation. The same also for the residuals, that were produced by including the homonymous function object in the `controlDict`. The drag force over the splitter body was then obtained from the `forces` function object.

Instead, the main flow features were visualized exploiting different filters from ParaView. Specifically, cutting planes and streamlines were used to analyze the behavior of the flow, iso-surfaces of the stoichiometric *air/fuel* ratio to study the development of the combustion.

4.3. Convergence

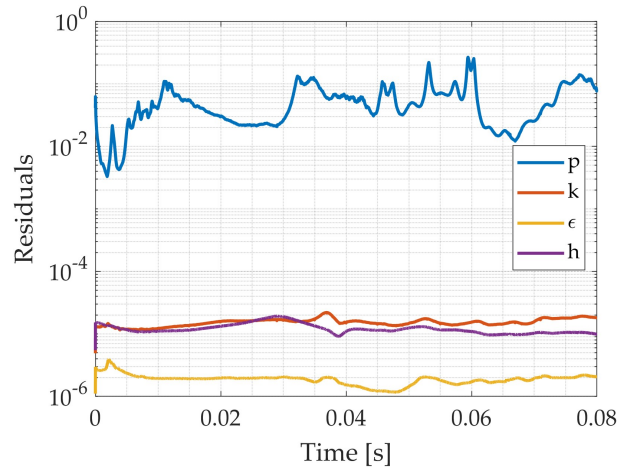


Figure 6: Evolution of the initial residuals over time

In Figure 6, the time evolution of the residuals for the main field variables during the hot-flow simulation is reported.

First, it can be observed that the residuals for the energy and turbulence transport equations show a very stable behavior and reach deep convergence, below 10^{-4} . This indicates that the activation of the combustion model and the associated strong heat release do not introduce significant numerical instabilities.

The pressure residual behavior is consistent with the findings from the precursor simulation, described in subsection 3.3. The residuals appear to stagnate at high values; however, as previously discussed, this is mostly related to the deactivation of the `momentumPredictor` in the

⁷ $SOI = 0$ s

PIMPLE algorithm. The variable density field induced by the combustion makes the initial guess for the pressure equation even more challenging compared to the cold-flow case. Nevertheless, the solver log confirms that the residuals drop significantly during the inner iterations of the PIMPLE loops, ensuring that mass conservation and velocity-pressure coupling are correctly resolved at each time step.

4.4. Results

4.4.1 Flow features

First, the phenomenology and the main flow features are analyzed from a qualitative perspective.

Figure 7 and Figure 8 illustrate the temporal evolution of the velocity field inside the combustion chamber.

In the upstream region, near the inlet, the effect of the Lagrangian spray of *n-Heptane* is clearly visible. In particular, at $t = 0.02$ s, the injection has just started, showing limited penetration. In the following time steps, as the spray becomes fully developed, the high velocity injection cone, driven by the high atomization pressure, can be clearly distinguished. Specifically, the velocity magnitude peaks at $t = 0.05$ s. This behavior is consistent with the half-sine injection flow rate profile described in subsection 4.1, which reaches its maximum intensity exactly at this instant. Toward the center of the domain, the V-shaped splitter acts as a bluff body. The flow is deviated around the obstacle, resulting in local accelerations. Immediately downstream, it is possible to observe the development of the wake region, characterized by near-zero velocities. This recirculation zone is further described by the streamlines analysis reported in Figure 9. It is also possible to observe the development of the boundary layer over the solid walls of the combustor. Finally, at $t = 0.08$ s, the high-momentum flow penetrates further downstream past the splitter, limiting the low-velocity recirculation region compared to previous time steps.

Notice that the results in the region immediately upstream of the `outlet` patch are likely affected by numerical diffusion, due to the coarse mesh resolution in that region.

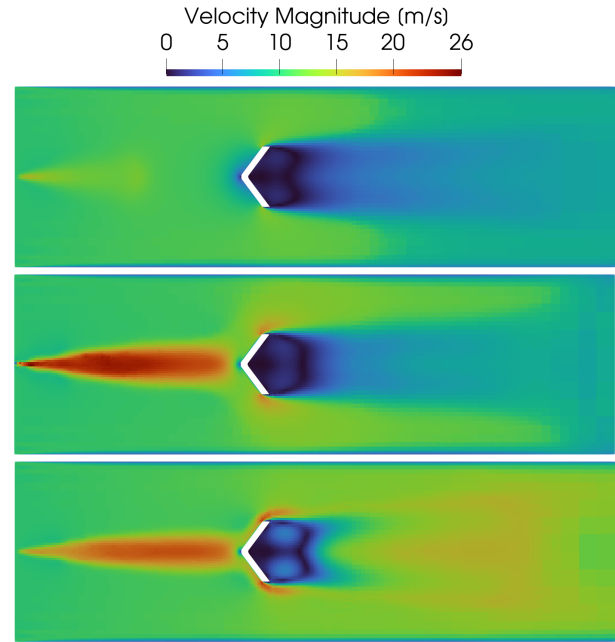


Figure 7: Velocity field in the xy symmetry cutting plane at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

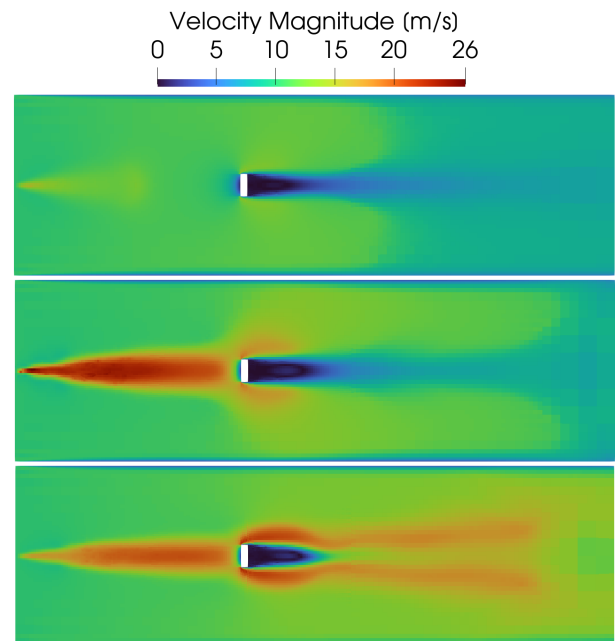


Figure 8: Velocity field in the yz symmetry cutting plane at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

Figure 10 and Figure 11 display the contours of vorticity magnitude in the xy and yz symmetry planes, respectively, at time instants immediately following the injection flow rate peak. The observed variations in the vorticity distribution are driven by the transient nature of the

injection profile.

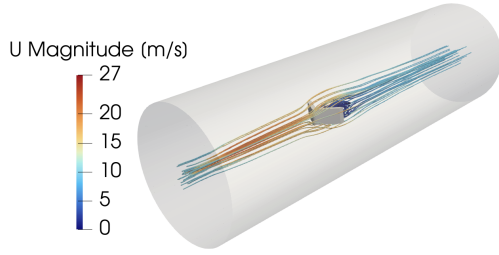


Figure 9: Streamlines visualization at $t = 0.05$ s

As the injection velocity follows a half-sine profile, the transition from the peak intensity ($t = 0.05$ s) to the deceleration phase induces a variation in the shear forces generated at the nozzle exit. This unsteadiness in the fuel injection contributes to the modulation of the vorticity structures entering the domain. The high vorticity magnitude observed upstream corresponds to the high-shear region generated by the high-velocity spray penetrating the slower moving air. This spray-induced turbulence, combined with the flow oscillation generated by the splitter, significantly enhances the mixing process. The interaction between the decelerating fuel jet and the bluff body wake creates a highly turbulent environment, characterized by the breakdown of large flow structures into smaller eddies. This phenomenon is clearly visible in the evolution of the vorticity contours between the showed time frames, confirming that the coupling of the transient spray and the bluff body aerodynamics is effectively in promoting the turbulent mixing of reactants.

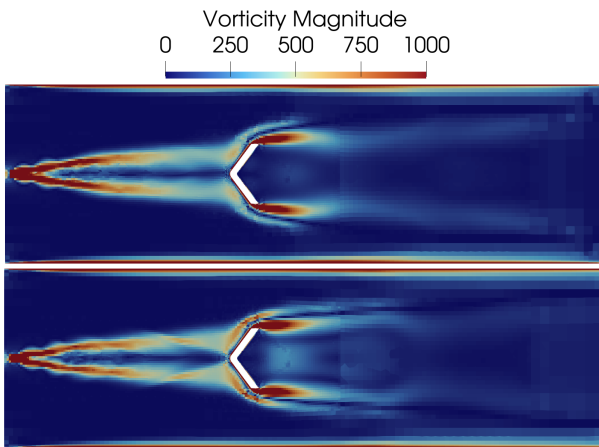


Figure 10: Vorticity distribution in the xy symmetry cutting plane at $t = [0.06; 0.07]$ s

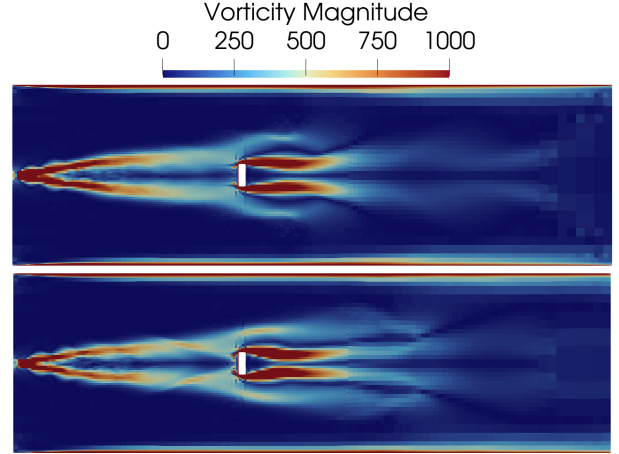


Figure 11: Vorticity distribution in the yz symmetry cutting plane at $t = [0.6; 0.07]$ s

In Figure 12 and Figure 13, the evolution of the temperature field on the xy and yz symmetry planes is shown. The sequence highlights the ignition and stabilization process of the *n-Heptane* combustion.

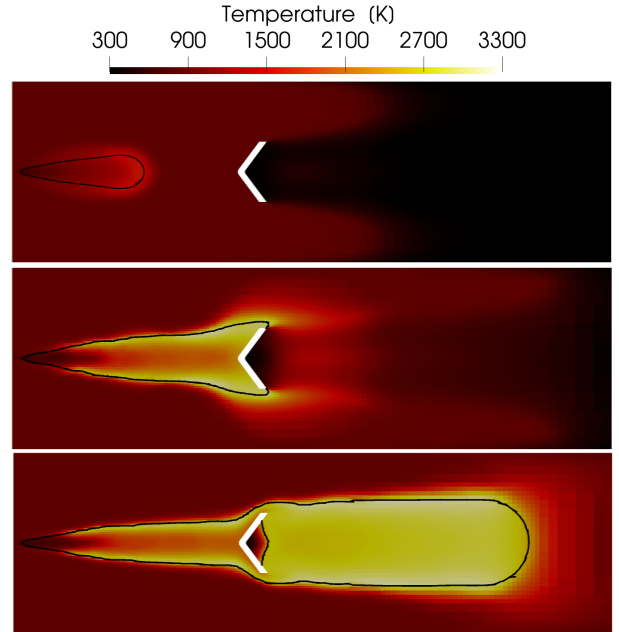


Figure 12: Temperature field in the xy symmetry cutting plane at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

Initially, at $t = 0.02$ s, only a fraction of the 95 mg of fuel has been injected into the combustor by the Lagrangian spray. Nevertheless, ignition has already occurred, leading to the formation of a flame front, identified by the stoichiometric *Air/Fuel* ratio iso-line marked in black. Since this represents the early stage of the simulation, the temperature field is not yet fully

developed.

As time progresses ($t = 0.05$ s), the hot gasses are transported downstream, forming a developed wake structure due to the interaction with the splitter. High temperatures can be observed; in particular, the distribution reflects the conical shape of the spray.

Finally, at $t = 0.08$ s, the temperature field appears stabilized, with the formation of a high-temperature core firmly anchored to the bluff body. This demonstrates the effectiveness of the recirculation zone in increasing residence time, thereby sustaining the flame against the high-velocity incoming flow and preventing blow-off.

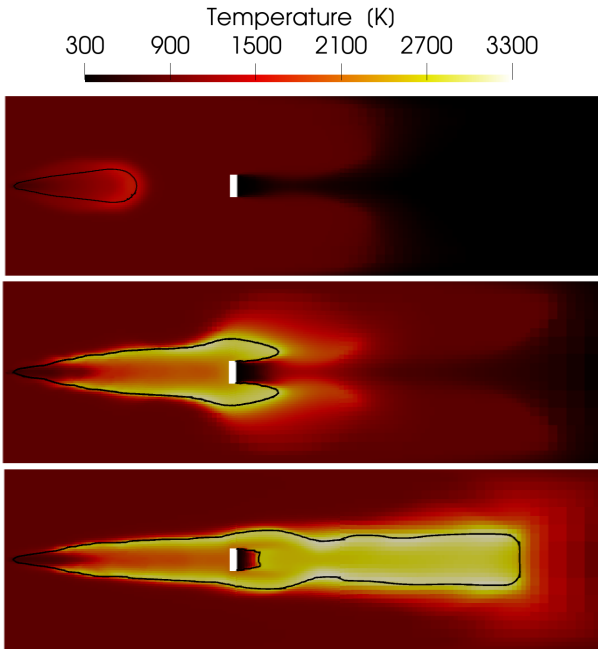


Figure 13: Temperature field in the yz symmetry cutting plane at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

The contour of the Mach number on the symmetry planes at $t = 0.05$ s is shown in Figure 14. This specific time instant was selected for the analysis because, as observed in the velocity field from Figure 7 and Figure 8, it corresponds to the moment of maximum velocity magnitude within the domain, driven by the peak of the spray injection profile.

The maximum value observed is approximately 0.05, localized in the high-velocity core of the spray injection upstream of the splitter. Instead, throughout the rest of the combustion chamber, and particularly in the wake region behind the bluff body, the Mach number remains extremely

low.

From a fluid dynamic perspective, this confirms that the flow is largely in the incompressible regime. Although a compressible solver is strictly necessary to handle the large density variations induced by the combustion of the mixture, the compressibility effects due to pressure are negligible. The density changes are therefore driven almost exclusively by the temperature gradients, rather than by pressure gradients.

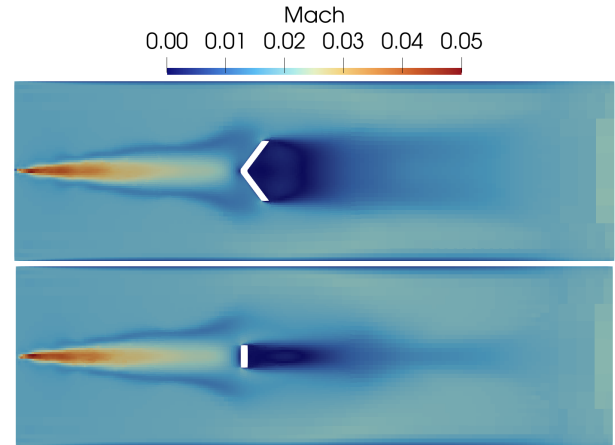


Figure 14: Mach number field at $t = 0.05$ s on the xy and yz symmetry planes

4.4.2 Combustion process

Now, a deeper insight into the combustion process is provided. While the temperature field gives a global indication of the heat release, the distribution of the chemical species allows for a precise identification of the flame front and the mixing efficiency.

Figure 15 presents the iso-surface⁸ of the stoichiometric *Air/Fuel* ratio, colored by temperature. Specifically, considering the single-step reaction employed for *n-Heptane*:



the stoichiometric condition corresponds to a mass ratio value equal to approximately 15.1.

This visualization effectively identifies the flame front. Consistent with the temperature cuts, the iso-surface elongates over time into a cylindrical shape. The high temperature on this surface

⁸Whose projection in the symmetry planes is shown in Figure 12 and Figure 13 by the iso-line marked in black.

confirms that the reaction is ignited where the mixture reaches a stoichiometric composition. The maximum temperatures observed approach the theoretical limits for the simplified chemical model employed⁹.

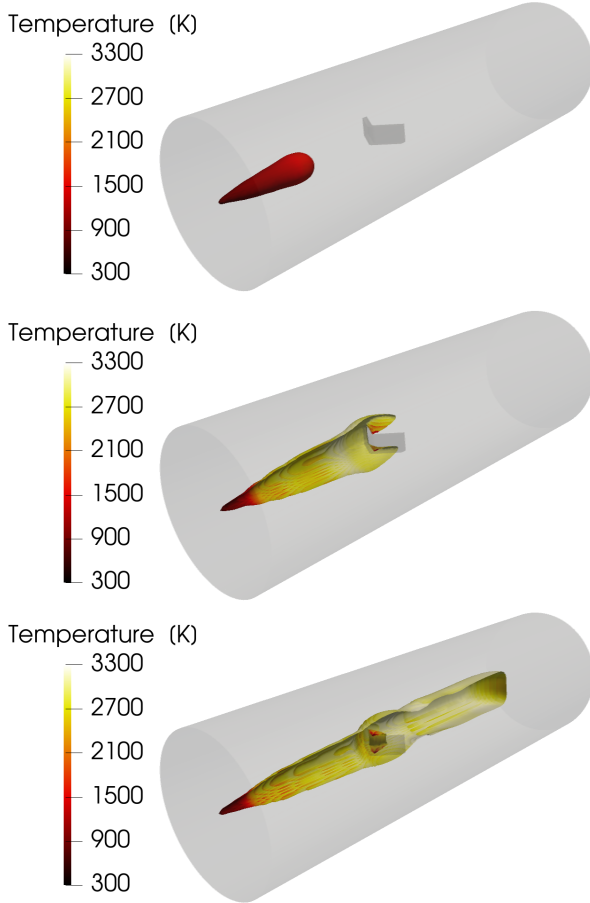


Figure 15: Iso-surface at stoichiometric *Air/Fuel* ratio at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

In order to gain a deeper insight into the flame structure and the combustion, the distribution in the symmetry planes of the main chemical species is analyzed at the time instant $t = 0.05$ s, corresponding to the peak of the injection phase. Figure 16 illustrates the mass fraction field of the oxygen, which represents the oxidizer in the chemical reaction (1). A distinct region of depletion (blue) is clearly observable in correspondence of the spray and in the splitter wake.

First, in the dense core of the spray immediately downstream of the injector, the droplets physically displace the air and starts evaporating,

⁹1-step combustion.

creating a mixture with lower oxygen concentration. Then, the oxygen is actively consumed by the chemical reaction. The gradient from the free-air stream to the reaction core highlights the mixture layer where the flame front is located.

Complementary to this, Figure 17 presents the distribution of carbon dioxide, which represents one of the main products of the reaction (1). As expected, the CO_2 concentration peaks in the regions corresponding to the maximum oxygen depletion, confirming that the reaction is successfully taking place.

The accumulation of the hot, reacted gas in front of the V-shaped splitter favors the continuous re-ignition of the fresh fuel-air mixture, as it provides the necessary thermal energy to sustain the flame against the high-velocity free stream. The comparison between these two visualizations confirms the non-premixed nature of the combustion; in fact, the reaction occurs in a conical structure where the fuel spray and the surrounding air interact, driven by turbulent mixing.

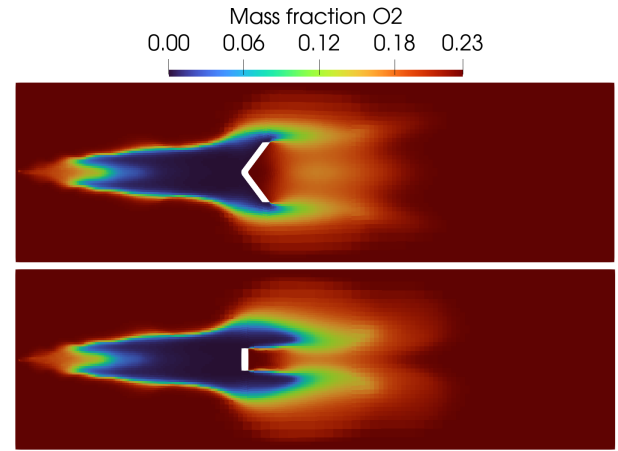


Figure 16: Oxygen mass fraction distribution at $t = 0.05$ s on the xy and yz symmetry planes

4.4.3 Lagrangian spray

To characterize the fuel injection process, the main features of the Lagrangian parcels are analyzed. The following visualizations provide a detailed view of the droplet size distribution and their penetration within the combustor.

Figure 18 illustrates the temporal evolution of the Lagrangian fuel particles, colored according to their diameter.

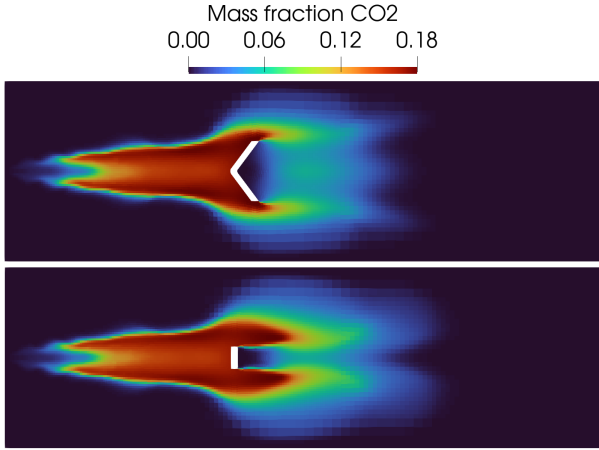


Figure 17: CO_2 mass fraction distribution at $t = 0.05$ s on the xy and yz symmetry planes

It can be observed that the particles diameter decreases significantly as the spray travels downstream. In fact, near the injector, larger droplets (red/orange) are dominant; however, as they move axially, they rapidly reduce in size (blue) and eventually disappear. This indicates an effective evaporation of the liquid phase, driven by the high temperatures of the surrounding reacting flow. As the combustion process develops over time, the rising temperatures further enhance the evaporation rate.

Furthermore, the spray cone remains well-confined within the central region of the combustor. Consequently, the liquid penetration length is relatively short compared to the domain length, and the spray angle is sufficiently narrow to prevent any impingement of the droplets onto the `outerWalls` and the splitter. The visualization confirms that the liquid fuel does not interact with the solid boundaries, because it fully evaporates before reaching them.

Thus, the simulation set-up described in subsection 4.1 is physically justified, confirming that wall-film modeling is not required.

To quantitatively assess the performance of the injection and breakup strategy (Reitz-Diwakar model), a statistical analysis of the Lagrangian parcels was performed. In particular, Figure 19 presents the histogram of the droplet diameter distribution at the instant of maximum injection intensity.

The spray exhibits a poly-disperse nature, with droplet diameters up to approximately $150\ \mu m$. The distribution follows a bell-shaped profile; specifically, the lower tail corresponds to the

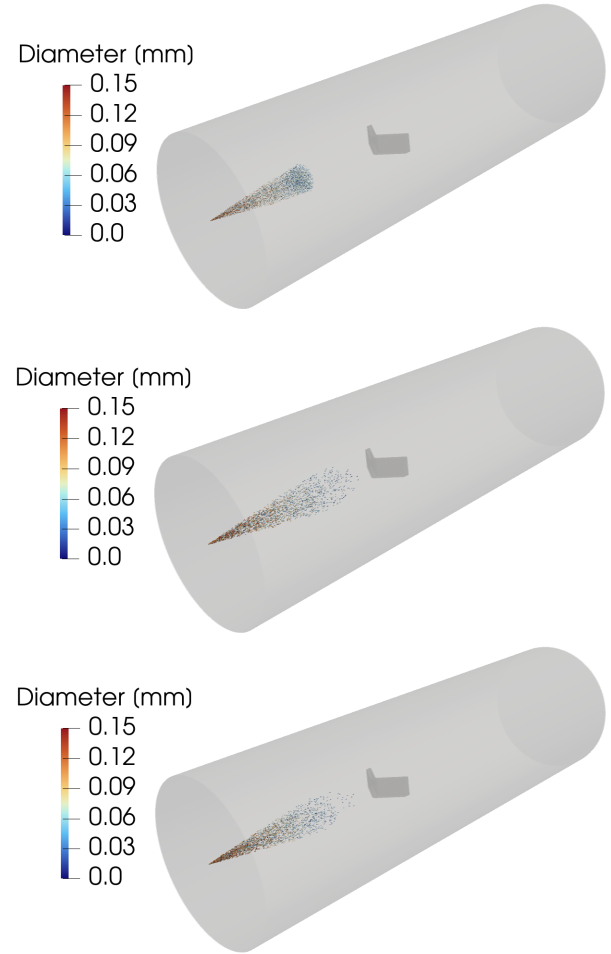


Figure 18: Fuel spray droplets at $t = [0.02; 0.05; 0.08]$ s (top, middle, bottom)

smallest droplets generated by the breakup mechanism or by the evaporation of larger parcels, while the larger droplets represent the liquid core particles that have not yet undergone secondary breakup.

A key parameter for characterizing the combustion is the Sauter Mean Diameter (SMD), indicated by the red line. This is defined as the diameter of a droplet whose ratio of volume surface area is equal to the volume-to-surface-area ratio of the entire distribution. At this time step, the value calculated is equal to approximately $105.22\ \mu m$. This quantity is of particular physical relevance because it emphasizes the total surface area of the liquid that is exposed to the heat transfer. In particular, a value in such range indicates that the spray provides a sufficient surface area for heat and mass transfer, leading to evaporation before the fuel reaches the domain boundaries, thus justifying the wall-film free approach previously discussed.

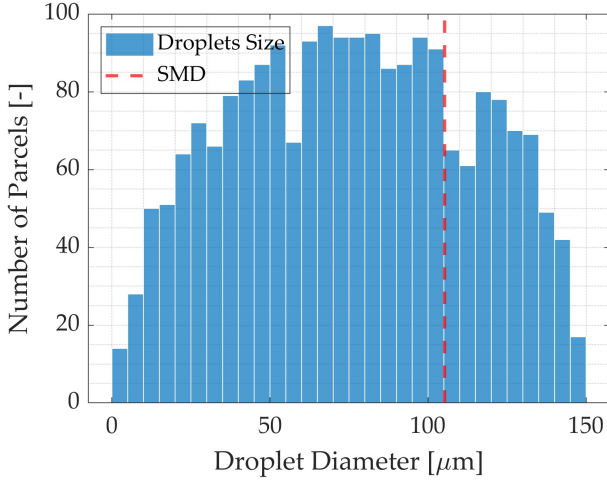


Figure 19: Histogram of the Lagrangian parcel diameters at $t = 0.05$ s

4.4.4 Quantitative analysis

Finally, a one-dimensional temporal and spatial analysis was performed to characterize the main flow features from a quantitative perspective.

The plot reported in Figure 20 shows the temporal evolution of the pressure averaged over the `inlet` and `outlet` of the combustor. Two distinct regimes can be identified.

In the initial phase of the simulation ($t < 0.04$ s), the pressure remains relatively stable around $1.015 \cdot 10^5$ Pa. This suggests that the heat released is initially negligible and insufficient to perturb the global pressure field. A minor transient is visible at the very beginning ($t < 0.005$ s), which can be attributed to the adjustment of the flow field upon activating the combustion and spray injection¹⁰.

After about 0.04 s, a sharp pressure rise can be observed. This represents a direct consequence of the combustion process, which releases a significant amount of heat in the domain. The pressure peaks at approximately 0.065 s, then it exhibits a slight decay towards the end of the simulation.

The `inlet` and `outlet` pressure traces are almost perfectly synchronized during the ramp-up, indicating that acoustic waves propagate rapidly throughout the combustor domain, effectively equalizing pressure gradients.

The evolution of the temperature averaged over

¹⁰Transition from the precursor cold-flow to the hot-flow run.

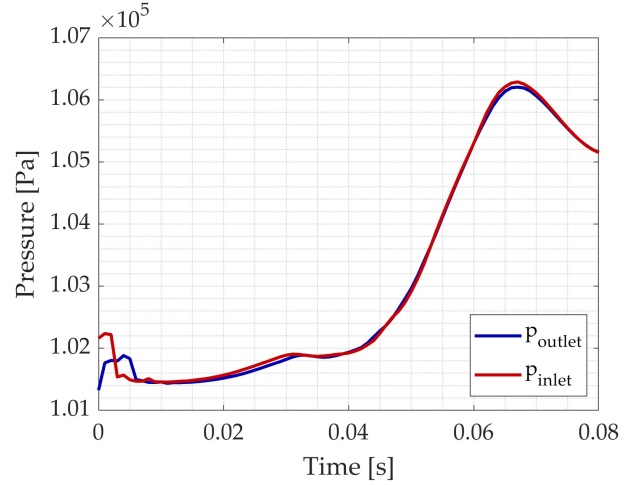


Figure 20: Evolution of the pressure averaged over the inlet and outlet patches

the `outlet` is illustrated in Figure 21.

The graph shows a constant value of approximately 300 K up to $t = 0.04$ s. This delay corresponds to the time required for the ignition of the combustion reaction and for the fluid particles to travel from the combustion zone to the outlet boundary.

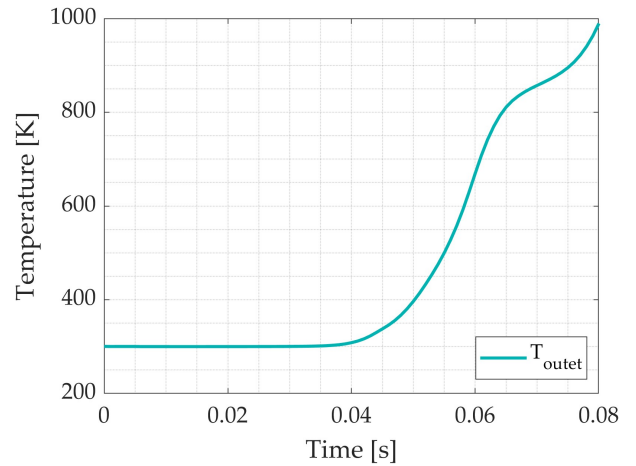


Figure 21: Evolution of the temperature averaged over the outlet patch

Once the hot products of the reaction reach the outlet patch, the temperature increases rapidly, consistent with the process described during the pressure analysis. By the end of the simulation, the average outlet temperature reaches approximately 1000 K. However, the slope of the curve at the final time step suggests that the thermal equilibrium (steady-state) has not been reached yet, and the mean temperature would probably continue to rise if the simulation was extended.

Notice that the temporal evolution of the temperature averaged over the `inlet` patch was not reported since the graph would have shown a constant value. In fact, the inlet temperature represents an imposed boundary condition, as described in subsubsection 3.1.1.

Figure 22 shows the temporal evolution of the total drag force exerted by the flow on the V-shaped splitter. Excluding the unphysical numerical spike at $t = 0$ s caused by the start of the simulation, the force time history reveals the intrinsically unsteady nature of the flow, driven by the coupling between the aerodynamics, the spray injection and the combustion. Two main regimes can be observed.

In the first phase of the simulation, the drag force fluctuates around a mean value of approximately $0.06 - 0.07$ N. Although the global pressure field is still relatively stable, as seen in Figure 20, the flow physics is complex. These oscillations are likely produced by the combined effect of the vortex shedding behind the bluff body and the momentum transfer from the developing fuel spray. In fact, the vaporized high-velocity fuel interacts with the wake, perturbing the recirculation zone.

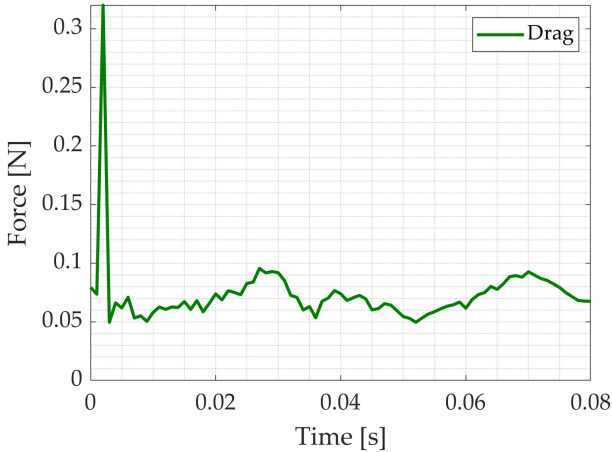


Figure 22: Evolution of the drag force on the splitter over time

Around $t = 0.04 - 0.05$ s, coincident with the steep pressure and temperature rise observed in the domain, the drag force exhibits a slightly increasing trend. This can be likely associated with the stabilization of the flame around the splitter; in fact, the volumetric expansion caused by the intense heat release accelerates the flow around the bluff body, modifying

the pressure distribution and contributing to increase the drag.

In Figure 23, the evolution of the y component of the velocity along a line centered in the domain and parallel to the y axis is shown. In particular, immediately downstream of the splitter, between $y \approx 0$ and $y \approx 0.08$ m, the velocity component becomes negative; in fact, this identifies the recirculation zone previously described. The negative velocity keeps the hot-products of the combustion process close to the splitter, contributing to the self-ignition of the fuel-air fresh mixture.

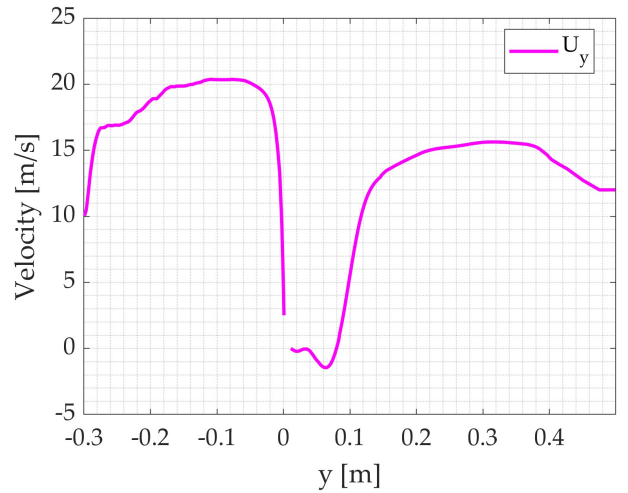


Figure 23: Evolution of the y velocity component over the y -axis at $t = 0.08$ s

Figure 24 reports the evolution of the temperature over the same centerline. Specifically, the peak exploited immediately after the splitter, from approximately 400 K to 2700 K, confirms that the flame is stably anchored in the wake. The maximum value exhibited, which is close to 2800 K, is coherent with the combustion model simulated. In fact, the simulation is adiabatic¹¹ and a simplified single-step reaction mechanism was considered; thus, it is reasonable to reach such high-temperatures.

Finally, in Figure 25, the evolution of the main chemical species involved in the reaction along the y -axis is illustrated. First, it can be underlined that the fuel mass fraction line is affected by fluctuations upstream of the splitter,

¹¹There is no heat dispersion through the solid walls due to the boundary conditions imposed in subsubsection 3.1.1.

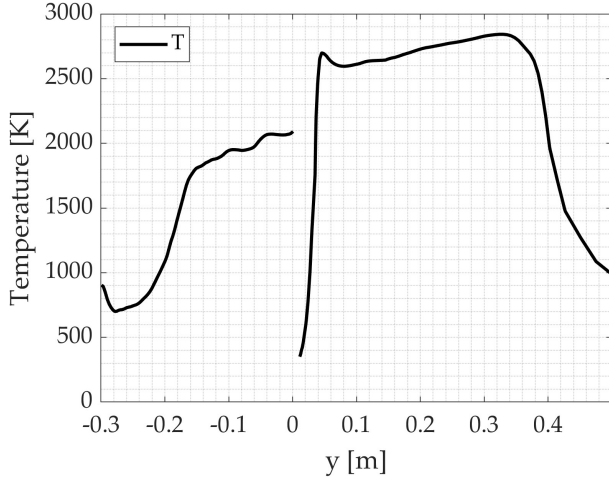


Figure 24: Evolution of the temperature over the y -axis at $t = 0.08$ s

which are due to the dispersed regime of the Lagrangian spray and the evaporation process. Then, the trend exhibited by the oxygen and CO_2 mass fractions is perfectly specular: when one rises, the other decreases.

Furthermore, at the end of the combustor, the fuel mass fraction becomes zero, indicating that all the injected fuel has been consumed by the combustion.

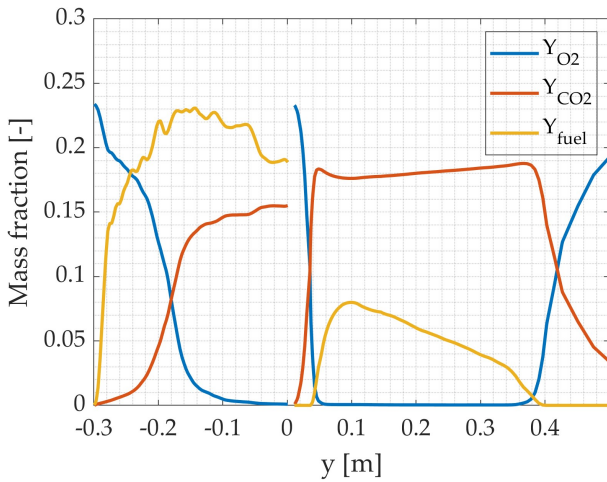


Figure 25: Evolution of the main chemical species along the y -axis at $t = 0.08$ s

Notice that the discontinuity in the distributions at $y \approx 0$ m is due to the presence of the splitter.

In order to quantitatively characterize the evolution of the flow field and the thermal mixing downstream of the splitter, transverse profiles of velocity magnitude and temperature

were extracted at two distinct axial locations¹²: $y = 0.05$ m, immediately downstream of the recirculation zone, and $y = 0.2$ m, further in the wake.

Figure 26 shows the recovery of the velocity deficit generated by the V-shaped bluff body. The profile at 0.05 m exhibits a pronounced "M-shape", which is characteristic of bluff-body wakes. Two high-velocity peaks are visible at $x \approx \pm 0.05$ m, corresponding to the accelerated shear layers detaching from the trailing edges of the splitter. In the central region, the velocity magnitude drops significantly, approaching zero. This velocity deficit confirms the presence of a strong recirculation zone where the flame is anchored.

Moving downstream to $y = 0.2$ m, the profile undergoes significant smoothing due to turbulent mixing. The momentum transfer from the high-velocity free-stream to the low-momentum region allows the wake to recover. Specifically, the peak velocities decrease, while the center-line velocity increases. This diffusive process is essential for stabilizing the combustion further downstream.

Notice that at the lateral boundaries, both velocities correctly drop to zero, satisfying the no-slip condition at the `outerWalls`.

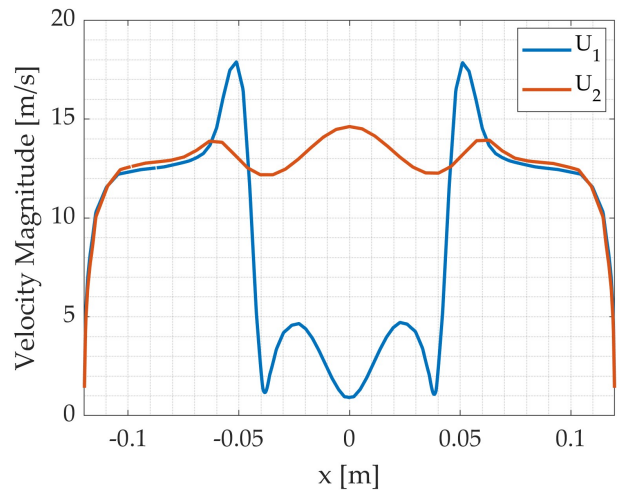


Figure 26: Evolution of the velocity magnitude over two transverse lines located at $y = 0.05$ m and 0.2 m at the final time step

The temperature distribution shown in Figure 27 provides insight into the flame structure

¹²The index 1 in the plots refers to the line at $y = 0.05$ m, while the index 2 identifies the other line.

and thermal diffusion.

In the near-wake region, the profile reveals a distinct high-temperature core confined within the splitter wake and limited by sharp gradients. This indicates that the combustion reaction is localized in the mixing layer between the fuel-rich recirculating flow and the oxidizer free-stream. Further downstream, the thermal profile widens. The steep gradients observed upstream are slightly smoothed out by turbulent heat transfer, distributing the thermal energy towards the combustion chamber walls. The persistence of high temperatures along the centerline confirms that the reaction is self-sustaining and that the residence time provided by the wake region is sufficient to achieve complete combustion of the injected fuel.

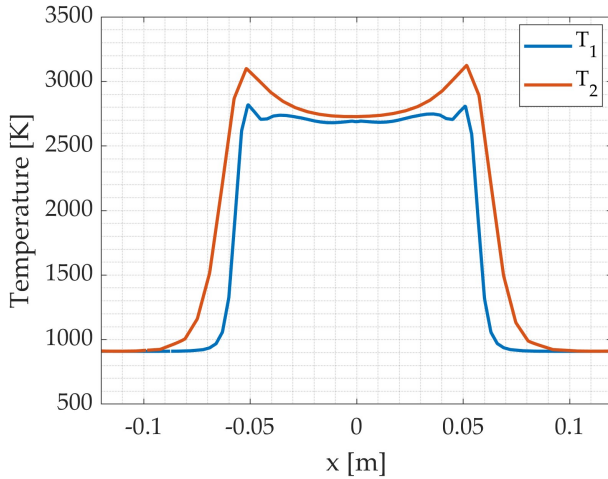


Figure 27: Evolution of the temperature over two transverse lines located at $y = 0.05\text{ m}$ and 0.2 m at the final time step

5. Conclusions

This work has presented the CFD analysis of a three-dimensional combustor in which liquid *n*-Heptane was injected into high-temperature air. The study was conducted with a preliminary cold-flow simulation to initialize the domain, followed by a reactive simulation that included Lagrangian spray.

The results obtained allow for a general understanding of the physical phenomena governing the problem. In particular, the analysis confirmed the effectiveness of the V-shaped splitter as a flame holder; in fact, the analysis demonstrated that the recirculation zone behind the

bluff body provides the necessary residence time and thermal energy to anchor the flame, preventing blow-off despite the high-velocity inflow. Furthermore, the Lagrangian particle tracking revealed a rapid evaporation of the fuel within the central region of the combustor without wall impingement.

However, it is important to acknowledge the limitations imposed by the computational resources available. While the current mesh is sufficient to capture the main features of the flow, it introduces numerical diffusion, particularly in the wake region and shear layers. A higher level of mesh refinement would be required to obtain more quantitatively accurate results. Specifically, a finer grid would allow for better resolution of the steep temperature and velocity gradients observed in the mixing layers and would reduce the numerical dissipation downstream of the splitter.

Nevertheless, the current simulation successfully highlights the fundamental coupling between aerodynamics, spray evaporation, and combustion.